Skybox Sampler

9 free stylized skies for Unity

Version 1.0.0 - The_KnightShift

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1. Overview

Skybox Sampler is a free package of 9 stylized skies for Unity: three space skies, three outdoor skies and three alien worlds, three from each of our three sky packs (Space Skyboxes, Stylized Sky Skyboxes and Alien Worlds Skyboxes).

Every sky is a Skybox/6 Sided material with its six 512x512 PNG cube faces already assigned, plus a lossless 2048x1024 PNG equirectangular panorama for Skybox/Panoramic or cubemap workflows. A demo scene and a small SkyboxCycler script let you flip through all 9 skies with the arrow keys.

All 52 skies of the three packs, with 1024x1024 cube faces, are in our paid package "Stylized Skybox Collection" on the Unity Asset Store (publisher The_KnightShift). This sampler uses the same folders and material names, so a sky you try here keeps its name there.

Quick start: open Demo/Demo.unity, press Play and use the Left / Right arrow keys to step through all 9 skies. Importing the package does not open the scene by itself.

Requirements: uploaded from Unity 6 (6000.0); older versions are not tested. The Built-in Render Pipeline uses the materials directly; for URP see section 5 (stated per release on the Asset Store page); HDRP uses a panorama through its HDRI Sky (section 5). No other packages are required.

2. Package contents

Everything is installed under Assets/TheKnightShift/SkyboxSampler, one folder per pack (Space, StylizedSky, AlienWorlds):

```
<Pack>/Materials/<Pack>_<sky>.mat          9 materials (Skybox/6 Sided)
<Pack>/Cubemaps/<sky>/{Front,Back,Left,Right,Up,Down}.png   6 x 512x512 per sky
<Pack>/Panoramas/<sky>.png                 9 panoramas, 2048x1024 PNG
Demo/Demo.unity                            demo scene with the SkyboxCycler
Scripts/SkyboxCycler.cs                    arrow-key sky switcher
                      (namespace TheKnightShift.SkyboxSampler)
Documentation/SkyboxSampler_Documentation.pdf   this document
README.txt, LICENSE.txt
```

Totals: 9 materials, 54 cube-face PNGs, 9 panorama PNGs, 1 scene, 1 script. All images are 8-bit sRGB (LDR) PNG files; there are no HDR/EXR files and no reflection probes.

3. The 9 skies and their packs

- 01 Teal & Magenta Nebula (Space Skyboxes) - emission gas in teal and magenta cut by dark dust lanes.
- 02 Ringed Gas Giant (Space Skyboxes) - a banded gas giant with a tilted ring system, lit from the side.
- 03 Supernova Remnant (Space Skyboxes) - a huge shell of filaments in teal, red and violet around a dying star.
- 04 Golden Hour (Stylized Sky Skyboxes) - low warm sun, glowing cumulus with bright rims and long shadows.

- 05 Thunderstorm (Stylized Sky Skyboxes) - towering dark storm clouds, a green-grey light and a forked lightning strike.
- 06 Moonlit Night (Stylized Sky Skyboxes) - navy sky, a bright moon with a halo, silver-edged clouds and stars.
- 07 Ringed Giant over Ocean (Alien Worlds Skyboxes) - a ringed gas giant fills the afternoon sky above a calm ocean that mirrors its rings.
- 08 Volcanic Ash World (Alien Worlds Skyboxes) - smothering ash clouds lit orange from below by rivers of lava, a smoking cone and cracked basalt underfoot.
- 09 Tidally-Locked Terminator (Alien Worlds Skyboxes) - a huge sun hangs half-risen on the horizon: a glowing day side melts into a star-filled night.

Each pack is in the Stylized Skybox Collection on the Unity Asset Store. The packs also have pages on itch.io:

```
Space Skyboxes         https://the-knightshift.itch.io/space-skyboxes
Stylized Sky Skyboxes  https://the-knightshift.itch.io/stylized-skyboxes
Alien Worlds Skyboxes  https://the-knightshift.itch.io/alien-skyboxes
```

Facing: the preset Rotation of each material

Each sky's main feature sits at its own bearing in the images. Every material's Rotation is preset so that the feature is in front of a camera looking along +Z, the default direction of a new Unity camera. The features are about 4 to 37 degrees above the horizon (Moonlit Night's moon is the highest); the demo camera looks 10 degrees up. Sky 9, Tidally-Locked Terminator, has files and a material named TidallyLockedTwilight. To face another way, change Rotation (0-360). A Skybox/Panoramic material made from a panorama needs the same Rotation for the same facing:

```
01 Teal & Magenta Nebula      Rotation   0
02 Ringed Gas Giant           Rotation   0
03 Supernova Remnant          Rotation   0
04 Golden Hour                Rotation  15
05 Thunderstorm               Rotation 348
06 Moonlit Night              Rotation 335
07 Ringed Giant over Ocean    Rotation 356
08 Volcanic Ash World         Rotation   0
09 Tidally-Locked Terminator  Rotation  55
```

4. Quick start (setup guide)

- 1. Import the package (Assets > Import Package > Custom Package, or from My Assets in the Package Manager). Keep everything selected and press Import.
- 2. Open Window > Rendering > Lighting and select the Environment tab.
- 3. Drag any material from a Materials folder (for example Space/Materials/Space_01_TealMagentaNebula.mat) into the Skybox Material slot. The sky appears in the Scene and Game views at once.
- 4. For lighting from the sky, set Environment Lighting > Source to Skybox and press Generate Lighting (or enable Auto Generate). Reflections follow the sky when Environment Reflections > Source is Skybox.

- 5. Optional: adjust the material's Exposure (brightness) and Rotation (turn the sky around the vertical axis; preset per sky, section 3) in the Inspector. Tint multiplies the colour.

To change the sky from code, set RenderSettings.skybox to any of the materials and call DynamicGI.UpdateEnvironment() so ambient lighting follows; SkyboxCycler.cs does exactly this.

5. Render pipelines

Built-in Render Pipeline

Use the materials directly as described in the quick start.

Universal Render Pipeline (URP)

URP renders Unity's built-in Skybox/6 Sided and Skybox/Panoramic shaders, so the materials are assigned in Lighting > Environment > Skybox Material exactly as in the Built-in pipeline, and the package has no custom shaders to convert. Whether this release was checked in URP is stated in the render pipeline table on its Asset Store page; if it is not listed there, treat URP as untested.

High Definition Render Pipeline (HDRP)

HDRP does not read the scene skybox material; it draws the sky from a Volume. Unity's documented route: select a panorama PNG, set Texture Shape to Cube, Mapping to Latitude-Longitude Layout (Cylindrical), apply, then add an HDRI Sky override to a Volume, assign the cubemap and set Visual Environment > Sky Type to HDRI Sky. The panoramas are LDR, so raise the HDRI Sky Exposure to taste. We have not tested this package in HDRP.

6. Demo scene

Open Demo/Demo.unity. It contains a camera tilted 10 degrees up, a directional light and a GameObject named SkyboxCycler whose Skyboxes array lists all 9 materials. Press Play, then use the Left and Right arrow keys to step through the skies; the current index and material name (which starts with the pack: Space_, StylizedSky_ or Alien_) are shown at the top left.

The scene has no meshes and no baked lighting, so it opens without a bake and nothing in it depends on the render pipeline. The cycler reads the keys through whichever input handling the project uses (Project Settings > Player > Active Input Handling: Input Manager (Old), Input System Package (New), or Both), so no setting needs changing.

7. Panoramas

Each sky's <Pack>/Panoramas/<sky>.png is a 2048x1024 equirectangular image already oriented for Unity's Skybox/Panoramic shader so that it matches the six cube faces of the same sky (with Rotation 0; give the Panoramic material the 6-sided material's Rotation for the same facing).

- Skybox/Panoramic material: create a material, choose shader Skybox/Panoramic, set Mapping to Latitude Longitude Layout and Image Type to 360 Degrees, and assign the PNG to Spherical (HDR).
- Cubemap asset: in the texture's import settings set Texture Shape to Cube and Mapping to Latitude-Longitude Layout; Unity builds the six faces for you (used by HDRP and reflection probes).

The panoramas are imported as plain 2D textures with Wrap Mode Repeat and mipmaps off (mipmaps produce a faint seam on lat-long images). They are lossless PNGs, so re-saving them is safe.

8. Script reference: SkyboxCycler

```
namespace TheKnightShift.SkyboxSampler
public class SkyboxCycler : MonoBehaviour
```

A small MonoBehaviour for previewing skies. Add it to any GameObject. Its own namespace means it does not clash with the SkyboxCycler of our other sky packages if you import several.

Public fields

- Material[] skyboxes - the materials to cycle through, in order. Fill it in the Inspector (lock the Inspector, select the materials in the Project window and drag them onto the array header).
- bool showLabel (default true) - shows '<index>/<count> <material name> (Left/Right arrow)' at the top left of the Game view with OnGUI.

Behaviour

- Start(): applies skyboxes[0] if the array is not empty.
- Update(): Right arrow applies the next material, Left arrow the previous one; wraps around at both ends.
- Apply(int i) (private): sets RenderSettings.skybox and calls DynamicGI.UpdateEnvironment() so ambient light follows the new sky. Copy this method if you want to switch skies from your own code.

Input: with the Input System package active (ENABLE_INPUT_SYSTEM) the script reads Keyboard.current; with the classic Input Manager (ENABLE_LEGACY_INPUT_MANAGER) it reads UnityEngine.Input; with Both it accepts either. It has no dependencies on other scripts or packages and can be deleted without affecting the materials.

9. Import settings

The package ships .meta files, so these settings are applied on import and nothing needs changing:

- Cube faces (54 PNGs): Texture Type Default, sRGB on, Wrap Mode Clamp, mipmaps off, Filter Bilinear, uncompressed (Compression None).
- Panoramas (9 PNGs): Texture Type Default, sRGB on, Wrap Mode Repeat, mipmaps off, Filter Bilinear, uncompressed.

To shrink builds, select the textures and set Compression to Normal Quality (or a platform override); Clamp wrap must stay on the cube faces or a thin line appears at the cube edges.

10. Troubleshooting

- Sky is blank or shows the default blue: the Skybox Material slot in Lighting > Environment is empty, or the Camera's Clear Flags is not Skybox.
- A thin seam at the cube edges: one of the six face textures lost its Clamp wrap mode; reset it.
- Sky looks too dark or bright: change Exposure on the material (the images are LDR, not HDR).
- The main feature of a sky is behind you: your camera does not look along +Z; change the material's Rotation (degrees around the vertical axis) or turn the camera.
- Arrow keys do nothing in the demo: make sure Demo/Demo.unity is the open scene, that Play mode is on and the Game view has focus (click it once).
- HDRP shows no sky: HDRP ignores the scene skybox material; follow section 5 (HDRI Sky).

11. License, support and AI disclosure

License: free. Your licence is the Unity Asset Store End User License Agreement (https://unity.com/legal/as-terms). See LICENSE.txt.

Support: use the support contact on this package's Asset Store page, or leave a comment on https://the-knightshift.itch.io/skybox-sampler. Please include your Unity version and render pipeline.

AI disclosure: these skies were generated by procedural Python code that was written with an AI coding assistant (Claude, by Anthropic). No photos and no image-generation model were used. The SkyboxCycler script, demo scene, material files, this documentation and the README were also written with AI assistance (Claude). The publisher opens the package in the Unity Editor and checks it by eye before release. The package contains no AI-generated audio or 3D models.